Atmospheric AI Take Home Assignment – Tom Maher

Question 1)

Part 1 – steam crackers

Assumptions:

1) All naphtha imported to a port is available to go to a cracker (none is blended or re-exported)
2) Active and operational facilities are both assumed to run at 80% of nameplate capacity (no distinction is made without turnaround data), unless a cracker's bid prices mean it buys less naphtha (see assumptions 9-10)
3) The midpoints of the ranges in the brief are used: a naphtha yield of 0.075 t per t of crude, and a naphtha consumption of 2.75 t per t of ethylene
4) An integrated complex is a facility with both a refinery and a cracker unit, or a refinery and a cracker of the same (parent) organisation within 3 km of each other. Its refinery supplies its cracker first, until the cracker reaches 80% of nameplate capacity
5) Pipelines between refineries/naphtha-importing ports and crackers are always used first (except in the case of integrated complexes), shortest first, even between different organisations. After that, where a (parent) organisation owns both refineries/naphtha-importing ports and crackers, its closest refinery/port-cracker pair is supplied first, then the next closest pair, until its crackers reach 80% of nameplate capacity or its refineries/ports run out of naphtha. Any naphtha or cracker requirement left over is dealt with under assumption 7
6) JVs are considered separate companies to their constituent partners
7) Naphtha remaining at ports and refineries goes to the crackers connected to them 1) via pipeline, then 2) via rail or barge, then 3) the geographically closest crackers, in that order, until it runs out
8) If a cracker's minimum bid is ≥ the ARA naphtha benchmark of 757 USD/t, or it has no bid on file, it takes its full naphtha requirement from any refinery or port
9) If the benchmark falls between a cracker's minimum and maximum bids, it buys only (maximum bid - 757) / (maximum bid - minimum bid) of its naphtha requirement on the market. Supply from refineries/ports of the same (parent) organisation, or by pipeline, is not limited
10) If a cracker's maximum bid is ≤ 757 USD/t, it buys no naphtha on the market (i.e. it is idle), UNLESS it has a refinery/port of the same (parent) organisation or a pipeline connection. Those still supply it, since sites under the same parent may trade on terms unrelated to the market price, and pipelines are assumed to always be used

What info, if any, is missing to give a complete picture of where the naphtha is going:

1) We have what a site is willing to pay for its feedstock but not what another site is willing to sell its product for, which will impact its choice of where to export
2) We don't have actual transport and purchase costs (e.g. demurrage fees, import/export duties, handling charges, FX costs, etc) or contractual pricing agreements (e.g. ex works, delivery duty paid, etc) to understand holistic costs for a cracker to purchase naphtha from a given refinery
3) Dynamic pricing info, i.e. how the feedstock bid prices and the ARA naphtha benchmark change at a given moment, which would influence purchasing decisions
4) Types of crude and crude slate into the refineries at a given moment (e.g. light sweet crudes such as Brent which contain a lot of light distillates and may yield more than 10% naphtha, compared to heavy sour crudes such as Venezuelan which would have a lower naphtha yield)
5) If a refinery's catalytic reforming unit is in a turnaround then that refinery will be exporting more naphtha as it can't convert it to higher value reformate (petrol). The operating status of units like this is important to know (i.e. more granular detail than simply is the refinery active or operational)
6) How much naphtha that a port imports ends up being blended and re-exported to give the net naphtha imported/exported to/from a port, and thus available to be transported to nearby crackers

Approach:

1) Calculate the actual throughput (ethylene production) of each cracker as its nameplate capacity x 0.8. Crackers with no listed design capacity are excluded
2) Calculate the naphtha output of each refinery as its nameplate capacity x 0.8 x 0.075. Refineries with no listed design capacity are excluded
3) Calculate the naphtha feed requirement of each cracker as its actual throughput x 2.75
4) Make a table of all naphtha-importing ports, their locations and the naphtha they imported in 2025
5) Make an empty output table with one row per cracker and supplying refinery/port. The columns are the cracker, the refinery/port, the transport mode, the naphtha supplied, and the share of the cracker's total naphtha supplied that comes from this refinery/port
6) Make a table of pipelines running from a refinery or naphtha-importing port to a cracker, with the (parent) organisations of both and the pipeline distance

7) Assign each facility a (parent) organisation. A facility with more than one direct owner belongs to that JV as its own company; facilities with the same owners belong to the same JV. Otherwise use the parent organisation id, or the direct owner where that is blank. Then make a table of organisations that own exactly one refinery/port AND one cracker, including integrated complexes
8) Make similar tables for organisations that own one refinery/port and multiple crackers, multiple refineries/ports and one cracker, and multiple refineries/ports and multiple crackers
9) Calculate the geographical distance between every cracker and every refinery/naphtha-importing port. For integrated complexes this is under 3 km, or zero where the refinery and cracker are the same facility
10) Calculate each cracker's market naphtha requirement as its naphtha feed requirement multiplied by:
    a. 1, if the minimum bid is ≥ 757 USD/t or there is no bid on file (flagged)
    b. (maximum bid − 757) / (maximum bid − minimum bid), if 757 USD/t is between the two bids
    c. 0, if the maximum bid is ≤ 757 USD/t
11) Keep a running balance of the naphtha remaining at each refinery/port, and of the feed requirement and market requirement remaining at each cracker. Update it after every allocation in steps 12–16. Each allocation is the smaller of the naphtha remaining at the refinery/port and the cracker's remaining feed requirement. If the refinery/port belongs to a different (parent) organisation and is not connected to the cracker by pipeline, the allocation is also limited by the cracker's remaining market requirement and is taken off it
12) Allocate naphtha from the refinery side to the cracker side of each integrated complex. An integrated complex is a facility with both a refinery and a cracker unit, or a refinery and a cracker of the same (parent) organisation within 3 km of each other
13) Allocate naphtha through the pipelines in step 6, shortest first
14) Allocate naphtha between facilities of the same (parent) organisation in steps 7 and 8, closest pair first
15) Allocate the naphtha remaining at each refinery/port to crackers connected to it by rail or barge, shortest first
16) Allocate the naphtha remaining at each refinery/port to any cracker, in order of distance, closest first
17) Label any feed requirement remaining at a cracker as:
    a. "Idle", if the cracker received no naphtha and its market requirement is zero
    b. "Reduced throughput", if its remaining market requirement is zero
    c. "Unallocated", otherwise
18) Label any naphtha remaining at a refinery/port as "surplus"
19) Fill the step 5 table with the allocations from steps 12–16. Each share is the naphtha supplied ÷ the total naphtha the cracker received. Calculate each cracker's actual throughput as the total naphtha received ÷ 2.75, and as a % of its nameplate capacity
20) Check that:
    a. Each supplied cracker's shares sum to 1
    b. No refinery/port supplies more naphtha than it has
    c. No cracker gets more than its market requirement from non-pipeline sources of other organisations
    d. Total refinery surplus ≥ total port naphtha exports, since imported naphtha is not re-exported, so exports must come from refineries
21) Rerun steps 2–19 with naphtha yield at 0.05 and 0.1 and naphtha consumption at 2.5 and 3, to see how sensitive each share is

See Python code 'Question 1.py' for the script that enacts this approach. A summary of a handful of facilities is given below.

Table 1. Naphtha suppliers of each cracker

Cracker	Cracker owner	Supplier	Supplier owner	Mode	Naphtha supplied (t/yr)	Share
Aliaga Petrochemical Complex	Petkim Petrokimya Holding A S (parent: SOCAR)	Aegean Aliaga (Star Ege) Refinery	SOCAR	integrated	639,812	0.716
Aliaga Petrochemical Complex	Petkim Petrokimya Holding A S (parent: SOCAR)	Api Ancona Refinery	Api Raffineria Di Ancona S p A (parent: SOCAR)	unrouted	254,131	0.284
Chemelot-Geleen Chemical Complex	SABIC Europe (parent: Saudi Basic Industries Corporation)	Holborn Refinery	Holborn Europa Raffinerie GmbH (parent: Oilinvest BV)	rail	202,932	1.000
Dunkirk Nord Petrochemicals	Versalis France SA (parent: Versalis SpA)	BP Rotterdam Refinery	BP Raffinaderij Rotterdam B.V. (parent: BP PLC)	pipeline	324,164	0.502
Dunkirk Nord Petrochemicals	Versalis France SA (parent: Versalis SpA)	Notre-Dame-de-Gravenchon Refinery	North Atlantic Energies	pipeline	321,392	0.498

Table 2. Status and actual throughput of each cracker

Cracker	Status	Naphtha feed requirement (t/yr)	Naphtha received (t/yr)	Actual throughput (t/yr ethylene)	% of nameplate
Aliaga Petrochemical Complex	Reduced throughput	1,143,597	893,942	325,070	62.5
Chemelot-Geleen Chemical Complex	Reduced throughput	1,209,459	202,932	73,794	13.4
Dunkirk Nord Petrochemicals	Reduced throughput	835,245	645,556	234,748	61.8

Here, unrouted means the transport type between facilities is unknown. To take one example, Chemelot: its nameplate capacity of 549,754 t of ethylene, at 80%, gives a feed requirement of 1,209,459 t/yr. The 757 USD/t price falls between its bids of 534 and 802 USD/t, so it will buy (802 - 757) ÷ (802 - 534) = 16.8% of its feed on the market, which is 202,932 t. It has pipelines from four refineries, but their naphtha had already gone to other crackers through shorter pipelines, so its supply comes via rail. Holborn Refinery's rail leg is listed at 14.1 km so Holborn supplies it. Holborn is a different company and not on a pipeline, so the sale is capped at the 202,932 t market requirement. That makes Holborn its only supplier, with a share of 1. The remaining 1,006,526 t is unmet, so Chemelot makes 73,794 t of ethylene, 13.4% of nameplate, and

is labelled reduced throughput. It is highly unlikely that a facility would run on 13.4% of nameplate (often that isn't technically possible due to minimum flow requirements on pumps and compressors etc), so evidently this simplified model is not accurate. The 14.1 km rail distance (not shown in this table) also looks wrong, since in reality the sites are about 400 km apart. This is an example of likely inaccurate data in the files. 25 of the 28 rail legs are shorter than the straight line distances. A similar issue occurs with pipelines in which 61 of 226 are listed as shorter than the straight line distance. In Chemelot's case it doesn't make a huge difference as its throughput is capped by its bid price range anyway, but the distances (and all data) would need to be sanity checked before running a more thorough sourcing model.

Part 2 - refineries

I did not have time to answer part two of this question. I would have taken a similar approach which was to outline what information I have available, write my assumptions, and then come up with a step-by-step approach for establishing how crude is routed from the ports to the refineries, and write a Python script that utilises this approach. As the ports do not have parent organisations, there would be no assumptions for the refinery sourcing model that are analogous to assumption 5 for the crackers. The primary means of determining where the crude is routed would thus be governed by the transport modes and distances between ports and refineries, as well as the bid price range per refinery and the crude benchmark price. To refine the model, it would be useful to know the amount and different types of crude being imported/exported through the ports, and the setup of the refineries (e.g. a hydroskimming refinery will only take light, sweet crudes, whereas complex refineries with a lot of HDS capacity will primarily take heavy, sour crudes). Benchmark prices more specific to each refinery configuration could thus be used to improve the accuracy of the model, as there are different benchmarks for the different types of crude (Brent, WTI, Dubai, Urals, etc).

Question 2)

Since there is no ground truth, I can't score a new version against real sourcing data. What I can do is run the old and new versions side by side on the same input data and compare them. This is similar to a management of change process that I was involved in when working at the Pernis refinery – we would run scenarios for proposed process changes whilst looking at P&IDs to understand possible adverse outcomes. Only once the effects were understood would we accept the change, and it would be monitored closely for a period afterwards to ensure the change was safe. In this case, I would still use the checks for step 20 in my approach.
For each cracker I'd check how much each supplier's share changes, and whether its main supplier changes. Changes should only appear where I'd expect them. For example, if I change the rule for rail and barge, crackers fed only by pipeline shouldn't move. Changes elsewhere probably mean a bug. I'd study the changes to the naphtha routings as well; a new version that suddenly sends naphtha an additional 1,500 km by rail needs explaining. Additionally, I'd do sensitivity analysis by rerunning both versions with the input ranges from step 21 (naphtha yield 0.05-0.1, consumption 2.5-3). A version whose answers swing a lot with small changes to uncertain inputs needs scrutinising. Shutting down some refineries and seeing how the model re-routes the naphtha would also be telling if it is giving sensible results.
When measuring the above, I would withhold certain known relationships (e.g. the location of integrated complexes, or crackers with a known pipeline to a refinery) when running the model and see if the new version still links them correctly.
Some reasons I would refuse to ship a change would be:

- Any physical check fails (step 20)
- A known case that was right becomes wrong, or a result is physically unintuitive
- Crackers change that shouldn't be affected by the change, and I can't explain why
- Results become much more sensitive to inputs during sensitivity analysis, and I'm not sure why

Question 3)

Firstly, I'd be careful with the precision when reporting (e.g. I would not say a cracker gets 62.34% of its naphtha from a certain source). For this example, integers would seem the sensible level of precision, with ranges or error margins included in the reporting. I'd also show narrower ranges on sourcing information that I have more confidence in. For instance, there's more confidence that naphtha goes between refinery and cracker in an integrated complex than one hundreds of miles away by barge. Honesty in the uncertainty is also important; part of the naphtha feedstock to a cracker may be unknown or unaccounted for, and that should be shown. Physical links, known transport modes and verified sourcing data are all sources that give fairly confident answers. If a link between refinery and cracker depends on distance alone, or inputs are missing in the data files, then that gives not very confident answers.
To keep a user from over trusting any answers, I would state from what sources the naphtha breakdown had been inferred (location, transport link, ownership, bid prices, etc), give an explanation as to what makes it a confident or not a confident answer, and always present data as a range.